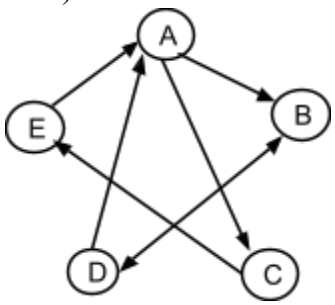
American Computer Science League

2023-2024 ● Contest 4: Short Problems ● Classroom Division

1. Graph Theory

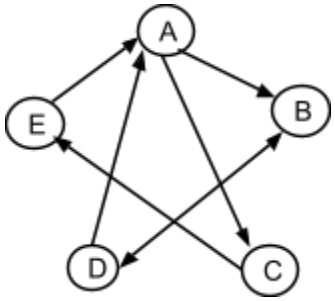
How many ordered pairs of vertices do not have paths of length 2 in the following directed graph? For example, (B,A) means from B to A. (This is the same graph as #2.)



A. 12
B. 14
C. 15
D. 16
E. 20

2. Graph Theory

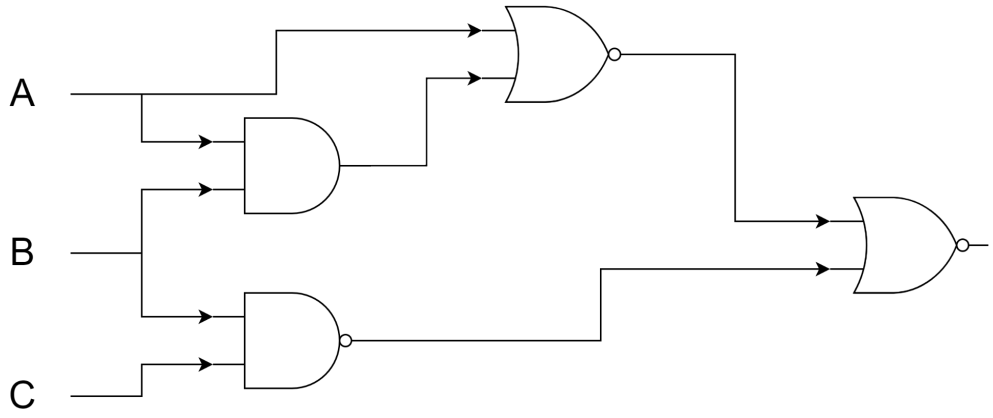
How many cycles are there in the directed graph below? (This is the same graph as #1.)



A. 2
B. 3
C. 4
D. 5
E. 6

3. Digital Electronics

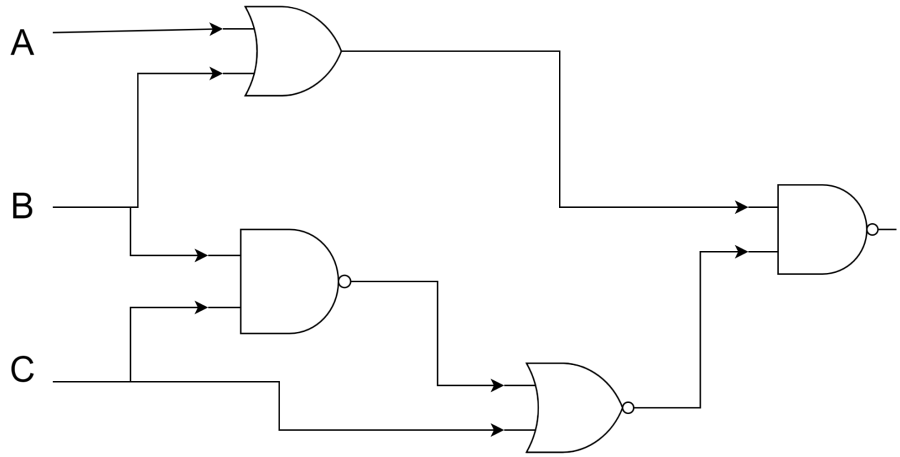
Simplify the Boolean expression represented by the following digital circuit:



A. $A + B + C$
B. $AB + C$
C. $A + BC$
D. ABC
E. BC

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. 8

5. Assembly Language

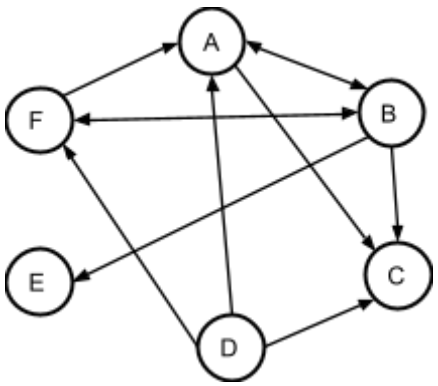
What is the output when the following program is executed?

```
T       DC      0
NUM     DC      28
        LOAD    =1
TOP     STORE   CNT
        LOAD    NUM
        DIV     =2
        SUB     CNT
        BL      STOP
        LOAD    NUM
        DIV     CNT
        MULT    CNT
        SUB     NUM
        BL      CONT
        LOAD    T
        ADD     CNT
        STORE   T
CONT    LOAD    CNT
        ADD     =1
        STORE   CNT
        BU      TOP
STOP    PRINT   T
        END
```

A. 10
B. 14
C. 28
D. 30
E. 56

6. Graph Theory

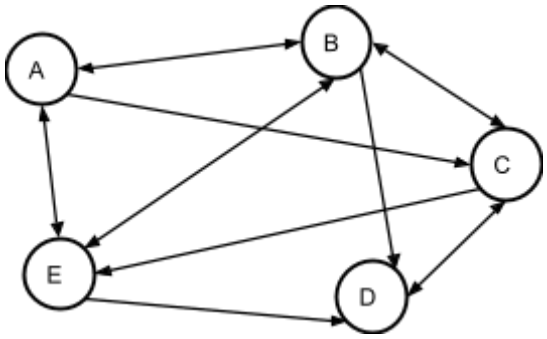
How many paths of length 2 are there in the following directed graph?



A. 12
B. 14
C. 15
D. 16
E. 18

7. Graph Theory

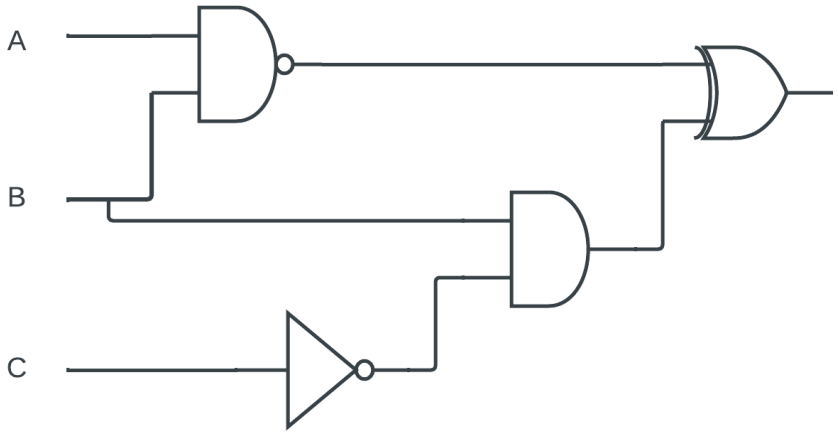
How many cycles are there in the following directed graph from vertex A?



A. 7
B. 8
C. 9
D. 11
E. 13

8. Digital Electronics

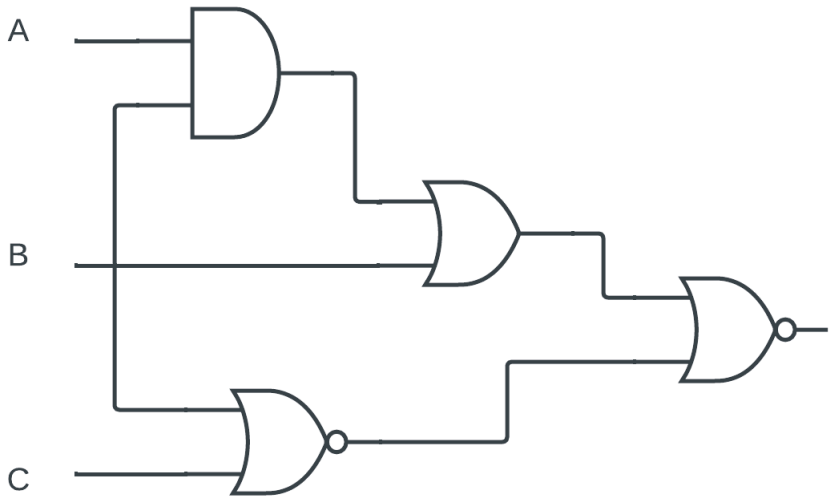
How many ordered triples make the following digital circuit TRUE?



A. 2
B. 4
C. 5
D. 6
E. 8

9. Digital Electronics

Simplify the Boolean expression represented by the digital diagram below:



A. $\overline{B}C$
B. $\overline{BC}$
C. $\overline{B} * \overline{C}$
D. $B\overline{C}$
E. BC

10. Assembly Language

What is the output when the following program is executed?

```
N         DC        1
T         DC        1
M         DC        78
          LOAD      T
TOP       SUB       M
          BE        STOP
          LOAD      N
          ADD       =1
          STORE     N
          LOAD      T
          ADD       N
          STORE     T
          BU        TOP
STOP      PRINT     N
          END
```

A. 11
B. 12
C. 13
D. 14
E. 15